

```
if (!isHigh) {
  highStartTime = millis();
  isHigh = true;
}
else if (millis() - highStartTime >= THRESHOLD_TIME) {
  isTripped = true;
  digitalWrite(RELAY_PIN, HIGH); // OPEN RELAY -> POWER CUT
  digitalWrite(GREEN_LED, LOW);
  digitalWrite(RED_LED, HIGH);
  digitalWrite(BUZZER_PIN, HIGH);

  lcd.setCursor(0, 0);
  lcd.print("ALERT: ILLEGAL! ");
  lcd.setCursor(0, 1);
  lcd.print("POWER TRIPPED  ");
}
}
else {
  isHigh = false;
}

delay(20);
}
```

User prompt: show everything step by step including all the connecting points on the breadboard

Response: Here is your exact breadboard coordinate map for the ESP32, I2C LCD, and 4-Channel Relay module.

1. Power & Shared Ground Rails

Source Component	Component Pin	Destination Breadboard Location	Notes
ESP32 Board	VIN or 5V	Red (+) Power Rail	Main 5V Logic Power
ESP32 Board	GND	Black (-) Ground Rail	Main System Common Ground
1.5V Battery	Negative (-) Wire	Black (-) Ground Rail	Common Ground
9V Battery	Negative (-) Wire	Black (-) Ground Rail	Common Ground

2. I2C LCD Module (HW-61) & Relay (HW-316) Logic

Module	Module Pin	Destination Location	Notes
I2C LCD	GND	Black (-) Ground Rail	Ground
	VCC	Red (+) Power Rail	5V Power
	SDA	ESP32 GPIO 21	Serial Data
	SCL	ESP32 GPIO 22	Serial Clock
Relay Module	GND	Black (-) Ground Rail	Ground
	VCC	Red (+) Power Rail	5V Power
	IN1	ESP32 GPIO 18	Relay 1 Trigger Signal

3. High-Voltage Relay Screw Terminals (K1)

Terminal Port	Wire / Connection Target	Role
K1 Center Port (COM)	9V Battery Positive (+) Red Wire	Direct 9V Power Input
K1 Left Port (NO)	Breadboard Column 20, Row e	Switched 9V Fence Output Line

4. Voltage Divider (3.0V ESP32 Protection Step-Down)

To protect the ESP32 ADC from exceeding 3.3V, three 10 kΩ resistors step 9V down to 3.0V.

Component	Terminal 1 Location	Terminal 2 Location	Role
10 kΩ Resistor #1	Column 20, Row d	Column 21, Row d	First half of R_1 (10 kΩ)
10 kΩ Resistor #2	Column 21, Row c	Column 22, Row c	Second half of R_1 (10 kΩ)
10 kΩ Resistor #3	Column 22, Row b	Black (-) Ground Rail	Ground resistor R_2 (10 kΩ)

📍 TOUCHPOINT 2 (Illegal 9V Fence Test Spot): Column 22 (Safe 3.0V output line).

5. Legal 1.5V Fence Input

Source Component	Wire Target Location	Role
1.5V Battery Positive (+)	Breadboard Column 15, Row e	Legal Power Line

📍 TOUCHPOINT 1 (Legal 1.5V Fence Test Spot): Column 15.

6. Indicator LEDs & Buzzer Grid Layout

Component	Pin / Terminal	Breadboard Location	Additional Connection
Blue LED (Fence #2 Live)	Anode (Long Leg)	Column 20, Row a	Connected to Relay NO (9V)
	Cathode (Short Leg)	Column 19, Row a	Bridge 220 Ω Resistor from Col 19, Row b → Black (-) Rail
Green LED (Authorized)	Anode (Long Leg)	Column 5, Row a	Jumper wire from ESP32 GPIO 16 → Col 5, Row b TX2
	Cathode (Short Leg)	Column 6, Row a	Bridge 220 Ω Resistor from Col 6, Row b → Black (-) Rail
Red LED (Trip Alarm)	Anode (Long Leg)	Column 9, Row a	Jumper wire from ESP32 GPIO 4 → Col 9, Row b
	Cathode (Short Leg)	Column 10, Row a	Bridge 220 Ω Resistor from Col 10, Row b → Black (-) Rail
Piezo Buzzer	Positive (+) Pin	Column 12, Row a	Jumper wire from ESP32 GPIO 19 → Col 12, Row b
	Negative (-) Pin	Black (-) Ground Rail	Direct Ground Connection

7. Portable Test Probe Wire

- Plug a Yellow Jumper Wire into ESP32 GPIO 34.
- Leave the other end loose:
 - Touch Column 15 to test Legal 1.5V Pass.
 - Touch Column 22 to test Illegal 9V Trip & Isolation.

User prompt: C:\Users\Lenovo\Downloads\fence_checker_esp32\fence_checker_esp32.ino:2:10: fatal error: LiquidCrystal_I2C.h: No such file or directory 2 | #include <LiquidCrystal_I2C.h> | ~~~~~ compilation terminated. exit status 1 Compilation error: LiquidCrystal_I2C.h: No such file or directory...this error is sowing

Response: This error occurs because the LiquidCrystal_I2C library is not yet installed in your Arduino IDE.

How to Install the Library in 30 Seconds:

1. Open Arduino IDE.
2. Go to Sketch → Include Library → Manage Libraries... (or press Ctrl + Shift + I / click the Library icon on the left sidebar).

The 3 External Wires Needed

- **Wire 1: Black Clip – Common Ground (GND)**
 - **Inside Box:** Connects to the main **Black (-) Ground Rail**.
 - **Outside Box:** Clips onto the ground/earth wire or negative terminal of the fence under test.
- **Wire 2: Yellow Clip – Voltage Sense Probe**
 - **Inside Box:** Connects to the **Input of your 3-Resistor Divider** (Breadboard Column 20, top leg of Resistor #1).
 - **Outside Box:** Clips onto the live fence wire to read its voltage.
- **Wire 3: Red Clip – Switched Relay Output**
 - **Inside Box:** Connects to the **Relay NO (Normally Open)** screw terminal.
 - **Outside Box:** Connects to the fence power line to cut off power if an illegal high voltage is detected.

Key Internal Circuit Modifications

1. Permanently Wire GPIO 34 to the Voltage Divider Output:

- Inside the box, keep the **GPIO 34 wire permanently plugged into Column 22** (between Resistor #2 and Resistor #3).
- Do not touch GPIO 34 directly to external wires anymore.

2. Feed External Signals Through the Divider Input:

- Run your external **Yellow Sense Wire** into **Column 20** (top of Resistor #1).
- This guarantees that any voltage (1.5V or 9V) entering through the Yellow Wire passes through the 3-resistor step-down before reaching GPIO 34, protecting the ESP32 permanently.

3. Power the Portable Box:

- Connect a 5V power bank or 9V battery (via a 5V buck converter / regulator) to the ESP32 **VIN** and **GND** pins so the box runs without USB power.
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To integrate the 1.5V battery as your legal fence signal, wire it into **Column 15** to act as your normal test point. [PDF](#)

Step-by-Step 1.5V Battery Setup

1. Connect Ground:

- Plug the **1.5V Battery Negative (-)** wire directly into the **Black (-) Ground Rail** on your breadboard. [PDF](#)

2. Connect Positive Terminal:

- Plug the **1.5V Battery Positive (+)** wire into **Breadboard Column 15**. [PDF](#)

3. Plug in GPIO 34 Probe:

- Connect your **GPIO 34 Yellow Probe Wire** directly into **Column 15**. [PDF](#)